dent Name (in Englis): _____ Student ID: _____

Question 1: Choose correct answer To be answered in (10) Minutes [7] / 8 Marks

1	2	3	4	5	6	7	8
D	B	D	B	D	D	A	A

1. Examples of $O(1)$ algorithms are:
   A. Multiplying two numbers
   B. Assigning some value to a variable
   C. Displaying some integer on console
   D. All of Above



2. The following array is a max heap: [10, 3, 5, 1, 4, 2].
   A. True
   B. False
3. The big O-notation of the expression $f(n) = 5n + 2^n + 6n^2$ is
   A. $O(n^2)$
   B. $O(n)$
   C. $O(1)$
   D. $O(2^n)$
4. Running merge sort on an array of size n which is already correctly sorted takes O(n) time
   A. True
   B. False
5. Which of the following is a stable sorting algorithm?
   A. Quicksort
   B. Merge sort
   C. Selection sort
   D. All of the above
6. Consider the recurrence: T(n)= T(n) + 1, using the Master method, choose the correct answer.
   A. Apply case 1 and $T(n) = \Theta(1)$
   B. Apply case 2 and $T(n) = \Theta(\log n)$
   C. Apply case 3 and $T(n) = \Theta(1)$
   D. Master method cannot be used.
7. Suppose you have the following recurrence relation $T(n) = 4T\left(\frac{n}{6}\right) + n$ then the num of levels in the tree are :
   A. $\log_6 n + 1$
   B. $\log_4 n + 1$
   C. $\log_2 n + 1$
8. Counting sort performs ..... Numbers of comparisons between input elements.
   A. 0
   B. n
   C. nlogn
   D. n^2

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Question 2: Time Complexity — To be answered in (20) Minutes — [8] / 8 Marks

1. **Consider the following running time function $T(n) = 10n^3 + 3n \log n$.**
   What is the big O-notation of $T(n)$?
   What are the closest constants c and n0? [2 marks]

$10n^3 + 3n\log n \leq 10n^3 + 3n^3$

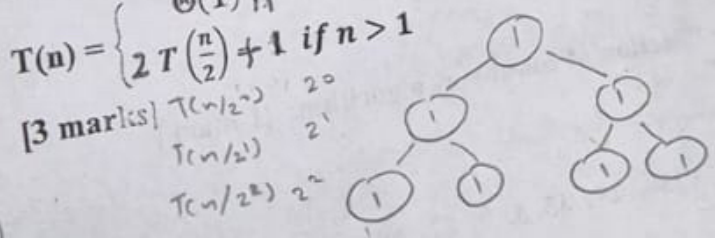
$10n^3 + 3n\log n \leq 13n^3$

$c = 13 \quad n_0 = 1$

2. **Use the recursion tree method to solve the questions for the following recurrence:**

$$T(n) = \begin{cases} \Theta(1) & \text{if } n = 1 \\ 2T\left(\frac{n}{2}\right) + 1 & \text{if } n > 1 \end{cases}$$

[3 marks]



$\Theta(n)$

$Total = \sum_{i=0}^{k} (2)^i + n = \frac{2^{\log n+1} - 1}{2-1} + n = 2 \cdot 2^{\log n} - 1 + n = 2n - 1 + n = 3n - 1 \rightarrow O(n)$

Put $n/2^k = 1 \Rightarrow k = \log n$

3. **Use the Master Method to get the Θ-notation for the following recurrence: [2 marks]**
   $T(n) = 8T(n/2) + n^3$

$a = 8 \quad b = 2 \quad f(n) = n^3$

$n^{\log_2 8} = n^3 = f(n) \Rightarrow$ case 2

$O(n^3 \log n)$

4. **What is the time complexity of the following algorithm in the most precise Big O? [1 mark]**

```
i ←1
while i<= n do
    j ←2
    while j<=i
        print j
        j = j *2
    endwhile
    i = i +1;
endwhile
```

$\log n$

n

$O(n \log n)$

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Question 3: Sorting Algorithm To be answered in (25) Minutes [9] / 9 Marks

a) How many loops in counting sort algorithm. Please specify each one with Big O. [3 marks]

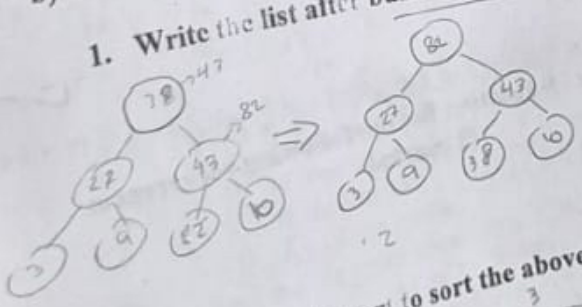
1) for i ← 0 to (k) → largest element in A
   → O(k)
2) for j ← 0 to (n) → A.length
   → O(n)
3) for i ← 2 to k
   → O(k)
4) for i ← n down to 0
   → O(n)

What is the recurrence function of count-sort algorithm. [1 mark]

$T(n) = 2T(n) + 2T(k)$ O(n)

b) Given the following list [38, 27, 43, 3, 9, 82, 10]

1. Write the list after building the max heap for the above list [2 mark]



82	27	43	3	9	38	10

2. Apply Radix sort to sort the above same list. [3 mark]

